

- Introduction to Malware and Malware Propagation Techniques
- Overview of Trojans, Their Types, and How to Infect Systems
- Overview of Viruses, Their Types, and How They Infect Files
- Introduction to Computer Worm



- Understanding the Malware Analysis Process
- Understanding Different Techniques to Detect Malware
- Malware Countermeasures
- Overview of Malware Penetration Testing





**Introduction
to Malware**



**Trojan
Concepts**



**Virus and Worm
Concepts**



**Malware Reverse
Engineering**



**Malware
Detection**



**Counter-
measures**



**Anti-Malware
Software**



**Penetration
Testing**

Malware is a malicious software that **damages or disables computer systems** and **gives limited or full control** of the systems to the malware creator for the purpose of theft or fraud

Examples of Malware

Trojan Horse

Virus

Backdoor

Worms

Rootkit

Spyware

Ransomware

Botnet

Adware

Crypter



Introduction to Malware



Trojan Concepts



Virus and Worm Concepts



Malware Reverse Engineering



Malware Detection



Counter- measures



Anti-Malware Software



Penetration Testing



Delete or replace **operating system's critical files**



Generate **fake traffic** to create DOS attacks



Record **screenshots, audio, and video** of victim's PC



Use victim's PC for **spamming and blasting email messages**



Download **spyware, adware, and malicious files**

Disable **firewalls and antivirus**



Create **backdoors** to gain remote access



Infect victim's PC as a **proxy server** for relaying attacks



Use victim's PC as a **botnet** to perform DDoS attacks



Steal information such as **passwords, security codes**, credit card information using keyloggers



Port	Trojan	Port	Trojan	Port	Trojan	Port	Trojan
2	Death	1492	FTP99CMP	5569	Robo-Hack	21544	GirlFriend 1.0, Beta-1.35
20	Senna Spy	1600	Shivka-Burka	6670-71	DeepThroat	22222	Prosiak
21	Blade Runner, Doly Trojan, Fore, Invisible FTP, WebEx, WinCrash	1807	SpySender	6969	GateCrasher, Priority	23456	Evil FTP, Ugly FTP
22	Shaft	1981	Shockrave	7000	Remote Grab	26274	Delta
23	Tiny Telnet Server	1999	BackDoor 1.00-1.03	7300-08	NetMonitor	30100-02	NetSphere 1.27a
25	Antigen, Email Password Sender, Terminator, WinPC, WinSpy,	2001	Trojan Cow	7789	ICKiller	31337-38	Back Orifice, DeepBO
31	Hackers Paradise	2023	Ripper	8787	BackOffice 2000	31339	NetSpy DK
80	Executor	2115	Bugs	9872-9875	Portal of Doom	31666	BOWhack
421	TCP Wrappers Trojan	2140	The Invasor	9989	iNi-Killer	33333	Prosiak
456	Hackers Paradise	2155	Illusion Mailer, Nirvana	10607	Coma 1.0.9	34324	BigGluck, TN
555	Ini-Killer, Phase Zero, Stealth Spy	3129	Masters Paradise	11000	Senna Spy	40412	The Spy
666	Satanz Backdoor	3150	The Invasor	11223	Progenic trojan	40421-26	Masters Paradise
1001	Silencer, WebEx	4092	WinCrash			47262	Delta
1011	Doly Trojan	4567	File Nail 1	12223	Hack'99 KeyLogger	50505	Sockets de Troie
1095-98	RAT	4590	ICQTrojan	12345-46	GabanBus, NetBus	50766	Fore
1170	Psyber Stream Server, Voice	5000	Bubbel	12361, 12362	Whack-a-mole	53001	Remote Windows Shutdown
1234	Ultors Trojan	5001	Sockets de Troie	16969	Priority	54321	SchoolBus .69-1.11
1243	SubSeven 1.0 – 1.8	5321	Firehotcker	20001	Millennium	61466	Telecommando
1245	VooDoo Doll	5400-02	Blade Runner	20034	NetBus 2.0, Beta-NetBus 2.01	65000	Devil

01

Create a new Trojan packet using a **Trojan Horse Construction Kit**

02

Create a **dropper**, which is a part in a trojanized packet that installs the **malicious code** on the target system



Attacker



Malicious Code

Example of a Dropper

Installation path: c:\windows\system32\svchosts.exe

Autostart: HKLM\Software\Mic... \run\Iexplorer.exe

Malicious code

Client address: client.attacker.com

Dropzone: dropzone.attacker.com

A genuine application

File name: chess.exe

Wrapper data: Executable file



Wrapper

03

Create a wrapper using **wrapper tools** to install Trojan on the victim's computer

04

Propagate the Trojan

05

Execute the dropper

06

Execute the damage routine





A wrapper **binds a Trojan executable** with an innocent looking .EXE application such as games or office applications



When the user runs the wrapped EXE, it first installs the **Trojan in the background** and then runs the wrapping application in the foreground



The two programs are **wrapped together** into a single file



Attackers might send a **birthday greeting** that will install a Trojan as the user watches, for example, a birthday cake dancing across the screen



✓ **Chess.exe**
File size: 90K



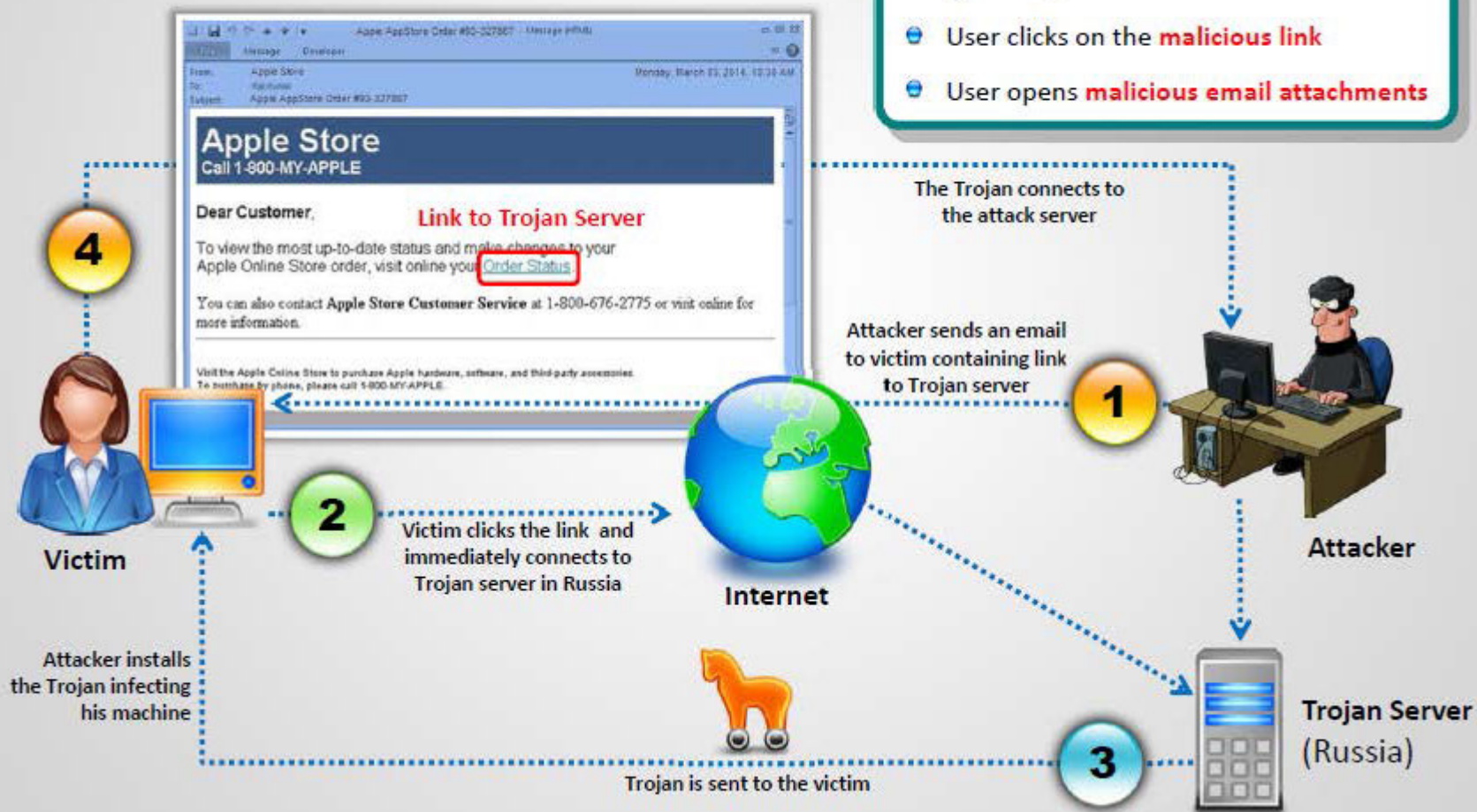
✗ **Trojan.exe**
File size: 20K



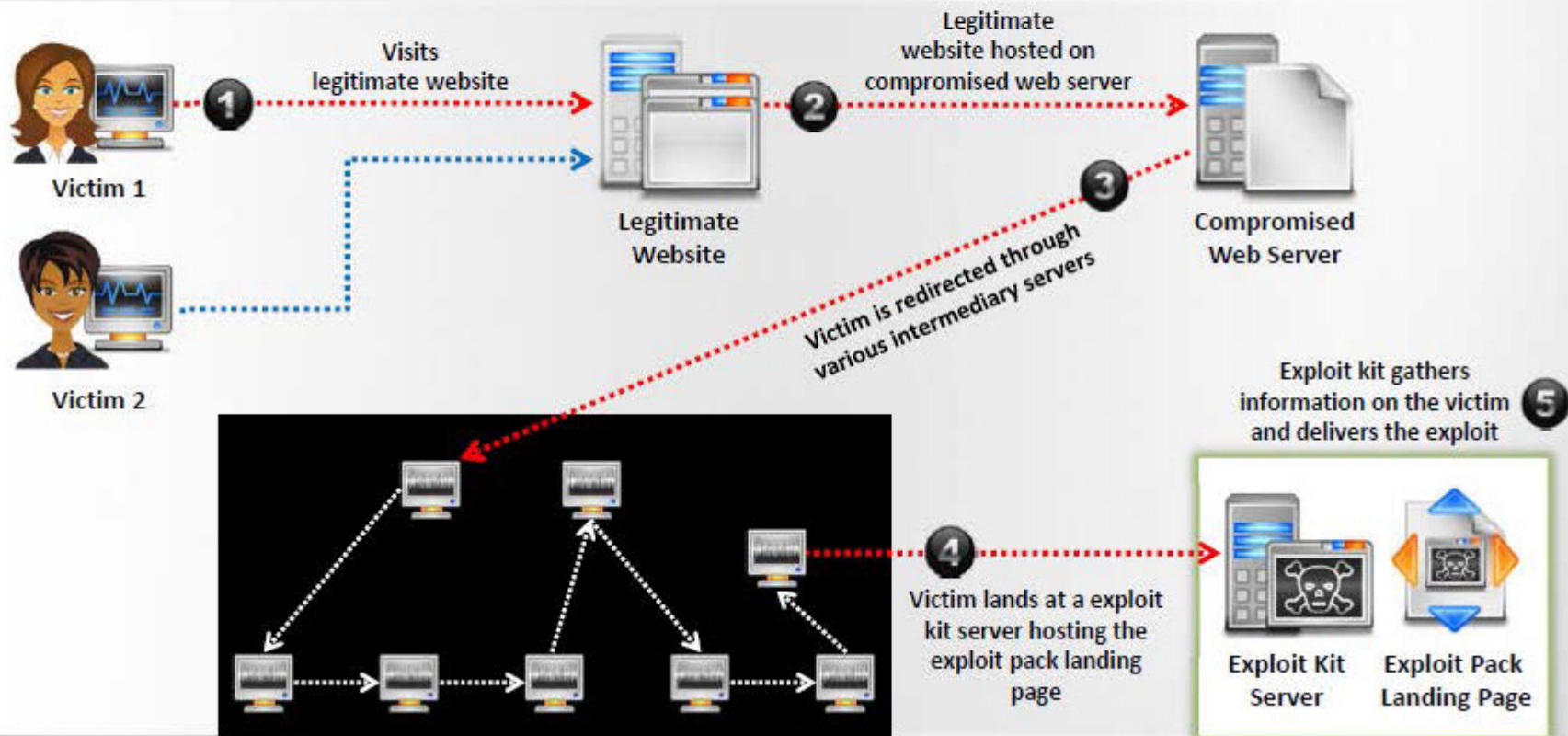
✗ **Chess.exe**
File size: 110K

Major Trojan Attack Paths:

- User clicks on the **malicious link**
- User opens **malicious email attachments**

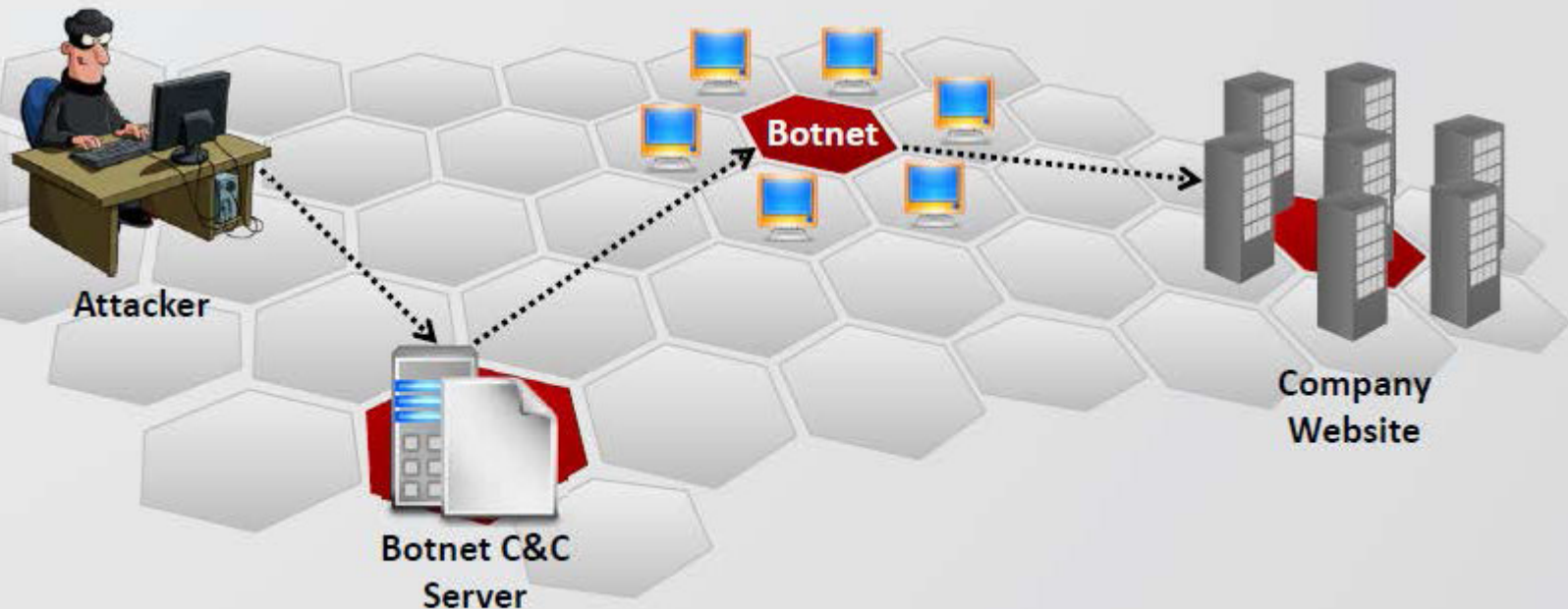


An exploit kit or crimeware toolkit is a platform to **deliver exploits and payloads** such as Trojans, spywares, backdoors, bots, buffer overflow scripts, etc. on the target system





- Botnet Trojans infect a large number of computers across a large geographical area to **create a network of bots** that is controlled through a Command and Control (C&C) center
- Botnet is used to **launch various attacks** on a victim including denial-of-service attacks, spamming, click fraud, and the theft of financial information



Proxy Trojan

Trojan Proxy is usually a standalone application that allows remote attackers to use the **victim's computer** as a proxy to connect to the Internet

Proxy server Trojan, when infected, starts a **hidden proxy server** on the victim's computer

Hidden Server

Infection

Thousands of **machines on the Internet** are infected with proxy servers using this technique



Attacker



Victim (Proxied)



Internet



Target
Company

Process

Encryption Trojan encrypts data files in victim's system and renders information unusable

"Your computer caught our software while browsing illegal porn pages, all your documents, text files, databases in the folder

My Documents was encrypted with complex password."



Attackers demand a ransom or force victims to make purchases from their online drug stores in return for the password to unlock files

"Do not try to search for a program that encrypted your information – it simply does not exist in your hard disk anymore," pay us the money to unlock the password



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Penetration Testing

- A virus is a **self-replicating program** that produces its own copy by attaching itself to another program, computer boot sector or document
- Viruses are generally transmitted through **file downloads**, **infected disk/flash drives** and as **email attachments**



Virus Characteristics



Infests other program

Alters data



Transforms itself

Corrupts files and programs



Encrypts itself

Self-replication



1**Design**

Developing virus code using programming languages or construction kits

2**Replication**

Virus replicates for a period of time within the target system and then spreads itself

3**Launch**

It gets activated with the user performing certain actions such as running an infected program

4**Detection**

A virus is identified as threat infecting target systems

5**Incorporation**

Antivirus software developers assimilate defenses against the virus

6**Elimination**

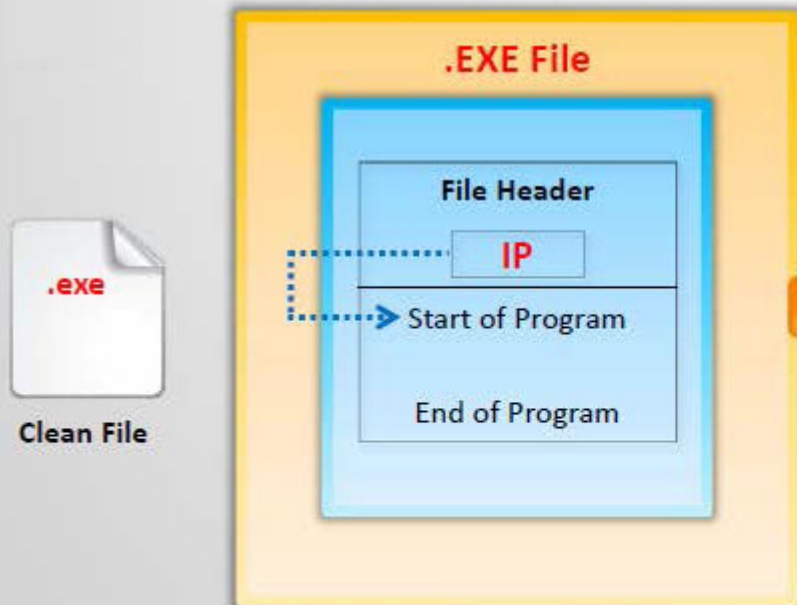
Users install antivirus updates and eliminate the virus threats



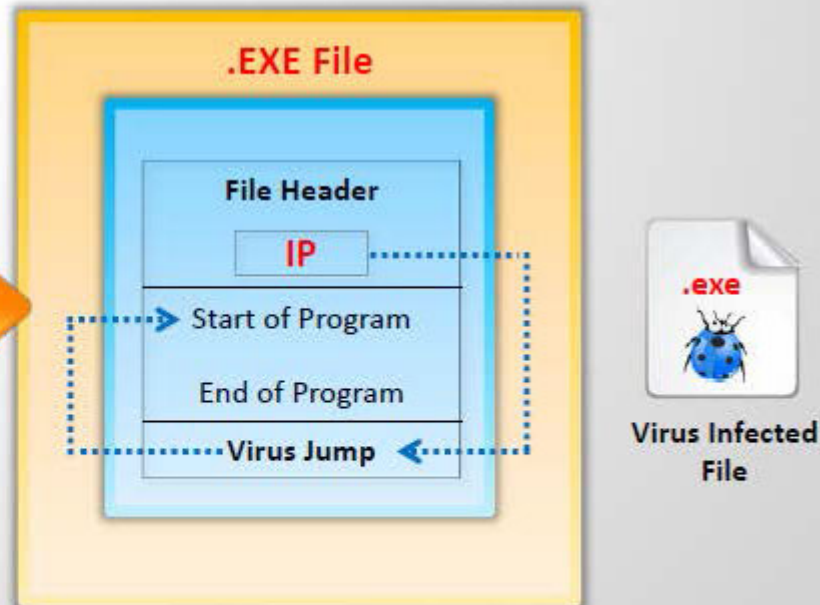
Infection Phase

- In the infection phase, the virus **replicates itself** and attaches to an .exe file in the system

Before Infection



After Infection



1



Inflict damage
to competitors



2



Financial
benefits

3



Research
projects

4



Play prank

5



Vandalism

6



Cyber
terrorism



7



Distribute
political messages

Processes take more resources and time

Computer beeps with no display

Drive label changes

Unable to load Operating system

Computer slows down when programs start



Anti-virus alerts

Computer freezes frequently or encounters error

Files and folders are missing

Hard drive is accessed often

Browser window "freezes"



Abnormal Activities

If the system acts in an unprecedented manner, you can suspect a virus attack



False Positives

However, not all glitches can be attributed to virus attacks





When a user accepts files and **downloads without checking** properly for the source



Opening **infected e-mail attachments**



Installing **pirated software**



Not updating and not installing new versions of **plug-ins**



Not running the latest **anti-virus application**

Ransomware is a type of a malware which **restricts access to the computer system's files and folders** and **demand an online ransom payment** to the malware creator(s) in order to remove the restrictions

Ransomware Family

- Cryptorbot Ransomware
- CryptoLocker Ransomware
- CryptoDefense Ransomware
- CryptoWall Ransomware
- Police-themed Ransomware

Your files are encrypted.
To get the key to decrypt files you have to pay **500 USD/EUR**. If payment is not made before **02/06/14 - 01:52** the cost of decrypting files will increase 2 times and will be **1000 USD/EUR**
Prior to increasing the amount left:
119h 57m 18s

Your system: Windows 7 (x32) First connected on [redacted] Total encrypted 38 files.

[Refresh](#) [Payment](#) [FAQ](#) [Decrypt 1 file for FREE](#) [Support](#)

We present a special software - CryptoWall Decrypter - which is allow to decrypt and return control to all your encrypted files.
How to buy CryptoWall decrypter?

bitcoin

1. You should register Bitcoin wallet ([click here for more information with pictures](#))
2. Purchasing Bitcoins - Although it's not yet easy to buy bitcoins, it's getting simpler every day.
Here are our recommendations:
 - [Coinbase](#) - Recommended for fast, simple service. Takes Credit Card, Debit Card, ACH, Wire
 - [LocalBitcoins.com](#) - Service allows you to search for people in your community willing to sell bitcoins to you directly
 - [coincore.com](#) - Another fast way to buy bitcoins
 - [bitquick.co](#) - Buy Bitcoins instantly for Cash
 - [How To Buy Bitcoins](#) - An international directory of bitcoin exchanges.
 - [Cash into Coins](#) - Bitcoin for cash.
 - [CoinJar](#) - CoinJar allows direct bitcoin purchases on their site.
 - [amoxa.com](#)
 - [bitflicious.com](#)
 - [Zinago](#) - Zinago is a global cash payment network enabling consumers to pay for digital currency.
3. Send 0.93 BTC to Bitcoin address: **1AK1xmw0Qd03Gp8B8H58KxZHQKJTVB** [Get QR code](#)
4. Enter the Transaction ID and select amount:
 0.93 BTC ~ 500 USD [Pay](#)
Note: Transaction ID - you can find in detailed info about transaction you made.
(example: 44214efca56ed332950c03929a430c4f19a27c426775d7c2aa06118cd412)
5. Please check the payment information and click "PAY".

PAY

Your sent drafts				
Name	Draft type	Draft number or transaction ID	Amount	Status
Your payments not found				

0 valid drafts are put, the total amount of 0 USD/EUR. The residue is 500 USD/EUR.

CryptoWall Ransomware

YOUR PERSONAL FILES ARE ENCRYPTED

All files including videos, photos and documents, etc on your computer are encrypted.

Encryption was produced using a **unique** public key generated for this computer. To decrypt files, you need to obtain the **private** key.

The single copy of the private key, which will allow you to decrypt the files, located on a secret server on the Internet; **the server will destroy the key after a time specified in this window.** After that, nobody and never will be able to restore files.

In order to decrypt the files, open site **4sfxtgpp53imlvzk.onion.to/index.php** and follow the instructions.

If **4sfxtgpp53imlvzk.onion.to** is not opening, please follow the steps below:

1. You must download and install this browser: <http://www.torproject.org/projects/torbrowser.html.en>
2. After installation, run the browser and enter the address: **4sfxtgpp53imlvzk.onion.to/index.php**
3. Follow the instructions on the web-site. We remind you that the sooner you do, the more chances are left to recover the files.



Police-themed Ransomware

Cryptorbit Ransomware



Remaining time: 47:58:43

MoneyPak MoneyGram



ATTENTION!

Your computer has been blocked up for safety reasons listed below.

You are accused of viewing/storage and/or dissemination of banned pornography (SMIL pornography/xenophilia etc); you have violated moral declaration on non-proliferation of child pornography; you are accused of committing the crime envisaged by Article 141 of United States of America criminal law.

Article 141 of United States of America criminal law provides for the punishment of deprivation of liberty for terms from 5 to 15 years.

Also, you are suspected of violation of 'Copyright and Related rights Law' (downloading of pirated music, video, warez) and of use and/or dissemination of copyrighted content; thus, you are suspected of violation of article 144 of United States of America criminal law.

Article 144 of United States of America criminal law provides for the punishment of deprivation of liberty for terms from 3 to 7 years or 150 to 550 basic amounts fine.

It was from your computer, that unauthorized access had been stolen to information of State importance and to data closed for public internet access.

Unauthorized access could have been arranged by yourself purposely on mercenary motives, or without your knowledge and consent, provided your computer could have been affected by malware. Consequently, you are suspected - until the investigation is held - of innocent infringement of article 215 of United States of America criminal law ('Law on negligent and reckless disregard of computers and computer aids').

Article 215 of United States of America criminal law provides for the punishment of deprivation of liberty for terms from 5 to 8 years and/or up to 100,000\$ fine.

Voucher NO: PPK Value

1 2 3 4 5 6 7 8 9 0

Pay MoneyPak Pay MoneyGram

How do I unlock the computer using the MoneyPak?

1. Find a retail location near you.
2. Look for a MoneyPak in the prepaid section. Take it to the cashier and load it with cash.
3. To pay fine you should enter the digits MoneyPak resulting into the payment form and press 'Pay MoneyPak'.



How do I unlock the computer using the MoneyGram xpress Packet?

1. Purchase a MoneyGram xpress Packet at a participating retailer.
2. Pick up a packet at one of the retailers listed below and send 1000 and 5000.
3. To pay fine you should enter the redemption number found inside your packet press 'Pay MoneyGram'.



- Boot sector virus **moves MBR to another location** on the hard disk and copies itself to the original location of MBR
- When system boots, **virus code is executed first** and then control is passed to original MBR

Before Infection



After Infection



File Viruses

- File viruses infect files which are **executed or interpreted in the system** such as COM, EXE, SYS, OVL, OBJ, PRG, MNU and BAT files
- File viruses can be either direct-action (non-resident) or memory-resident

Multipartite Virus

- Multipartite viruses infect the system **boot sector** and the **executable files** at the same time



Attacker





Macro viruses **infect files** created by Microsoft Word or Excel



Most macro viruses are written using **macro language Visual Basic for Applications (VBA)**



Macro viruses infect **templates** or **convert infected documents into template files**, while maintaining their appearance of ordinary document files



Attacker



.....→
Infects Macro Enabled Documents



User



File extension viruses **change the extensions** of files



.TXT is safe as it indicates a pure text file



With extensions turned off, if someone sends you a file named **BAD.TXT.VBS**, you will only see **BAD.TXT**



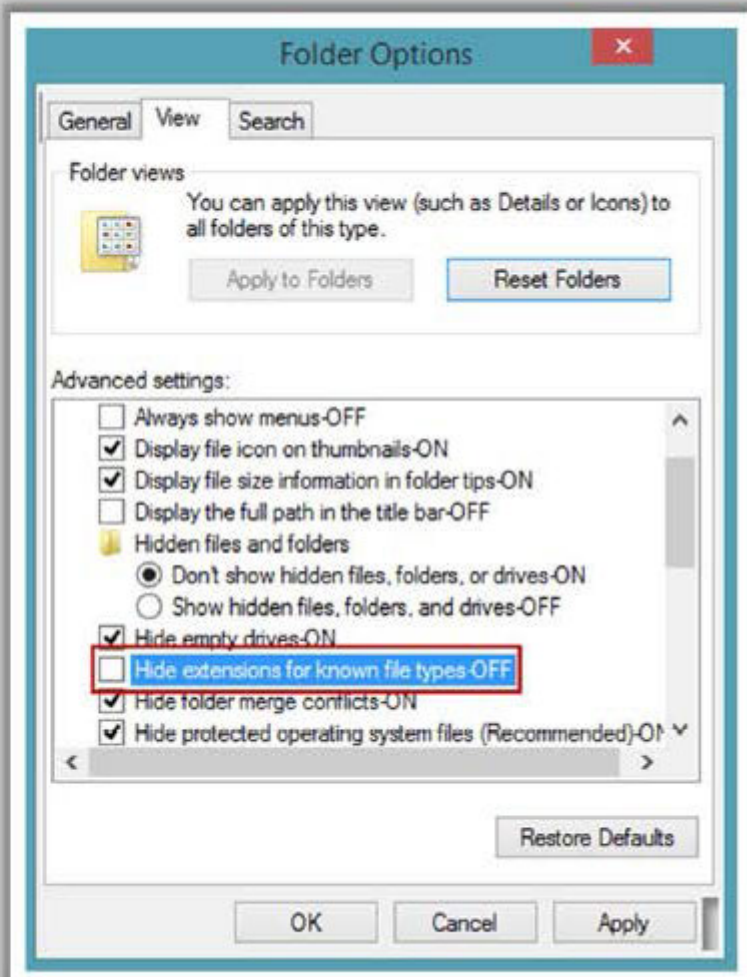
If you have forgotten that extensions are turned off, you might think this is a **text file** and open it



This is an **executable Visual Basic Script** virus file and could do serious damage



Countermeasure is to turn off "**Hide file extensions**" in Windows



Basic Infection Techniques

Direct Action or Transient Virus



- **Transfers** all the controls of the host code to where it **resides in the memory**
- The virus **runs when the host code is run** and terminates itself or exits memory as soon as the host code execution ends

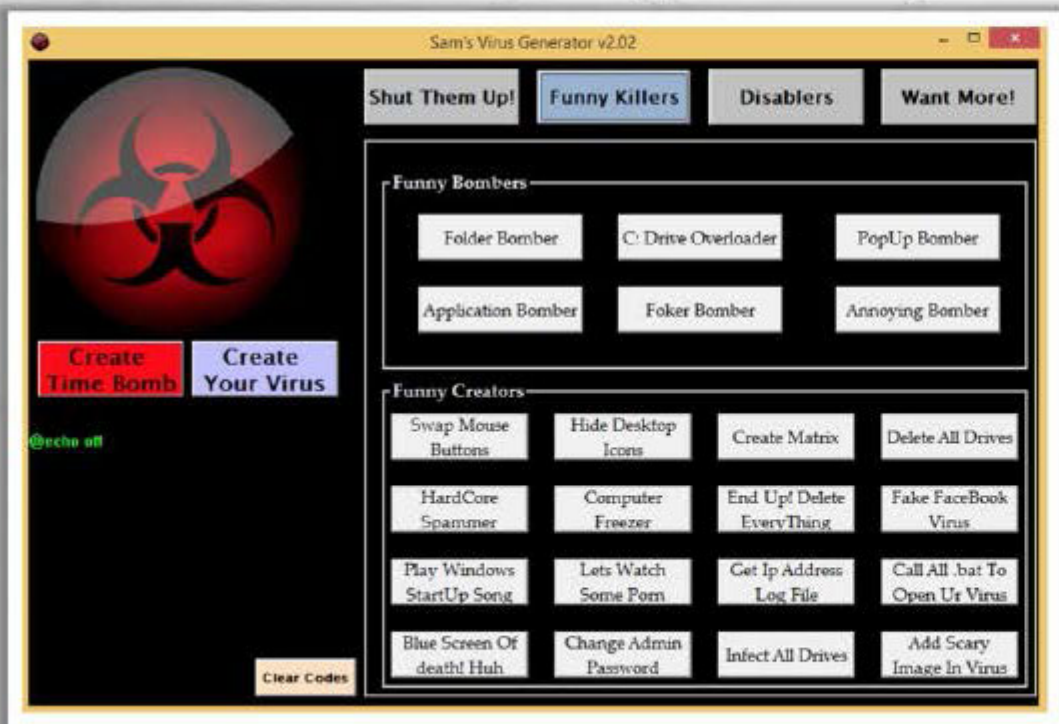
Terminate and Stay Resident Virus (TSR)



- **Remains permanently in the memory** during the entire work session even after the target host's program is executed and terminated; can be removed only by **rebooting the system**

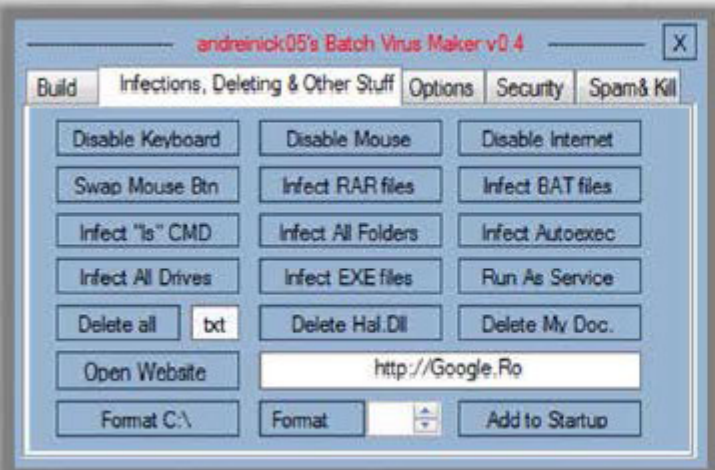


Sam's Virus Generator

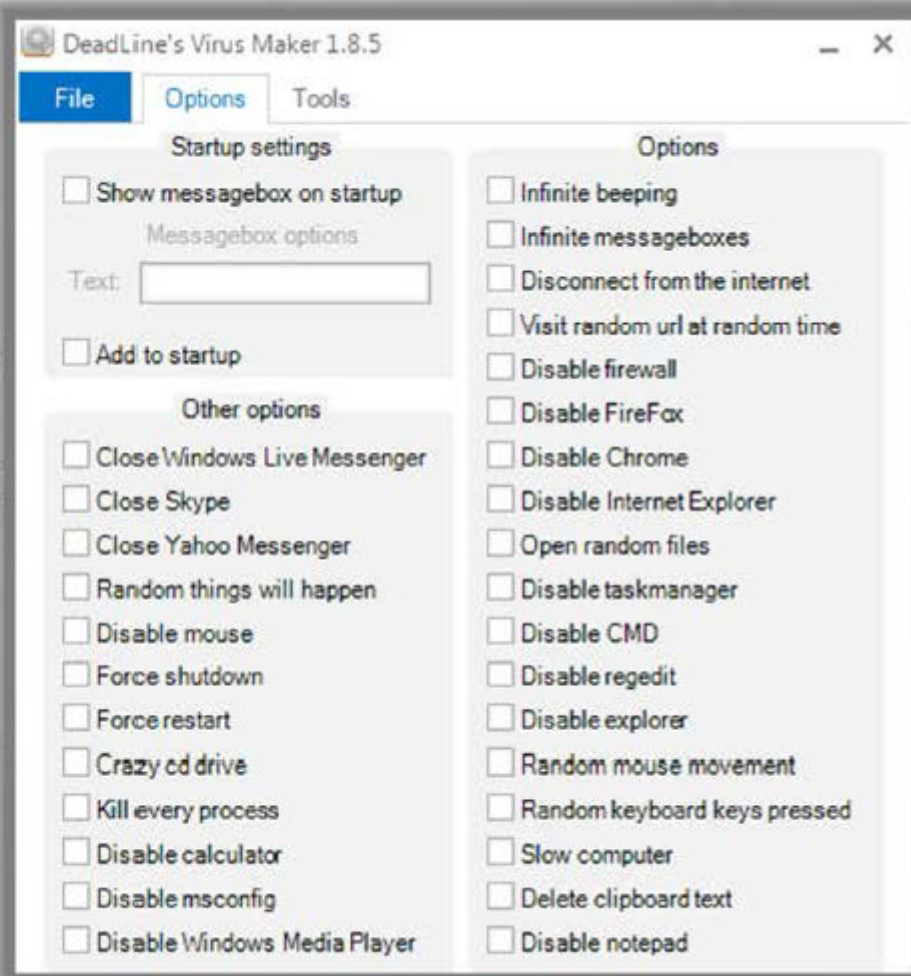


JPS Virus Maker





Andreinick05's Batch Virus Maker



DeadLine's Virus Maker

1

Computer worms are malicious programs that **replicate**, **execute**, and **spread** across the network connections independently **without human interaction**



Most of the worms are created only to replicate and spread across a network, consuming available computing resources; however, some worms carry a payload to **damage the host system**

2

3

Attackers use **worm payload** to install backdoors in infected computers, which turns them into zombies and **creates botnet**; these botnets can be used to carry further cyber attacks





Replicates on its own

A worm is a special type of malware that can replicate itself and **use memory**, but **cannot attach** itself to other programs



Spreads through the Infected Network



A worm takes advantage of **file** or **information** transport features on computer systems and spreads through the **infected network** automatically but a virus does not



- Anti-virus sensor system is a collection of computer software that **detects and analyzes malicious code threats** such as viruses, worms, and Trojans. They are used along with sheep dip computers



Network



Anti-Virus System

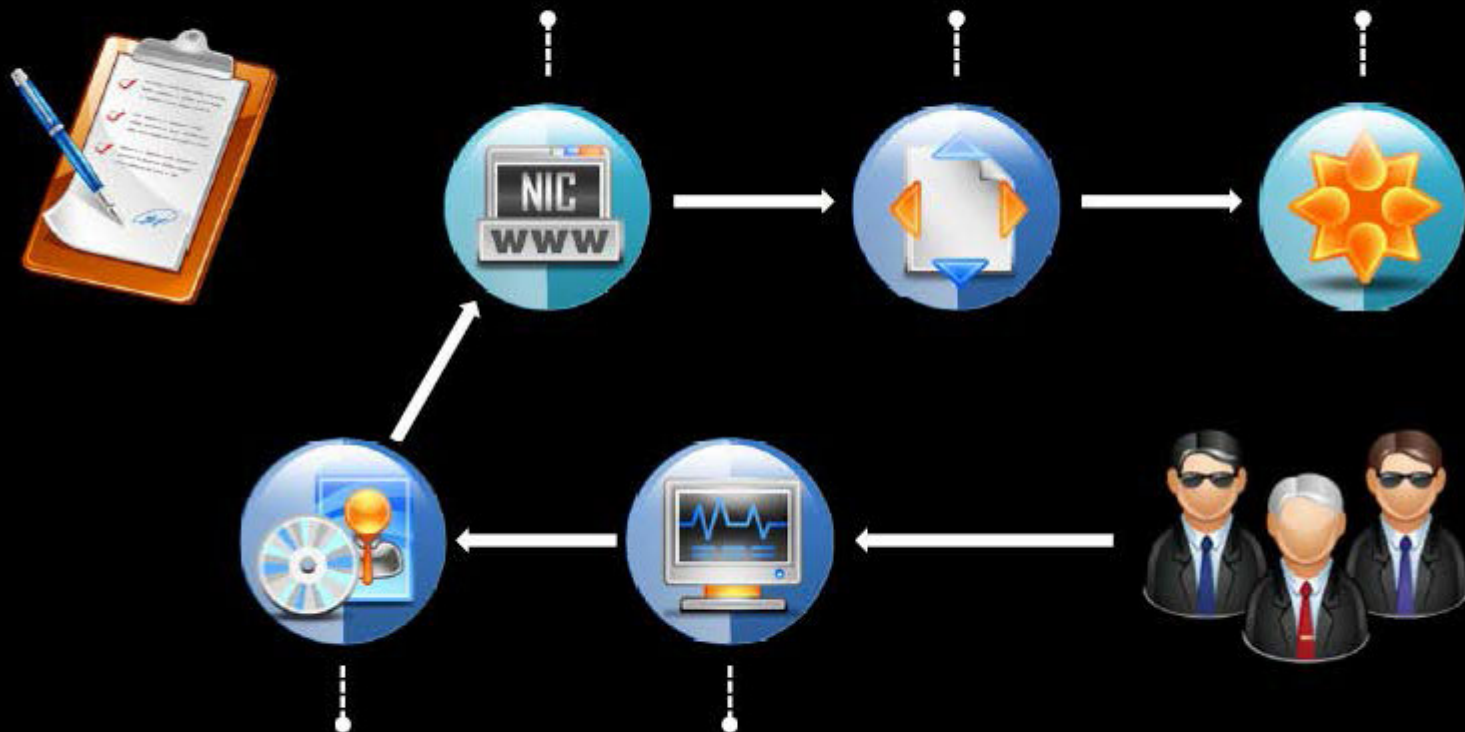


Internet

Isolate the system from the network by ensuring that the **NIC card** is in "host only" mode

Disable the '**shared folders**', and the '**guest isolation**'

Copy the **malware** over to the guest OS



Install **guest OS** into the Virtual machine

Install **Virtual machine** (VMware, Hyper-V, etc.) on the system

- VirusTotal is a free service that **analyzes suspicious files and URLs** and facilitates the detection of viruses, worms, Trojans, etc.



<http://www.virustotal.com>

A screenshot of the VirusTotal analysis results page. The page shows the following information:
- SHA256: ee29e80a2e8c469655fe215eac14c2fbb201116e40fd056dcd1f602e1959263b
- File name: pwdump7.zip
- Detection ratio: 37 / 40
- Analysis date: 2014-03-11 13:46:14 UTC (1 day, 19 hours ago)
On the right side, there is a circular progress indicator showing 37 out of 40 detections, with a red arrow pointing to the 37 and a green arrow pointing to the 40.
Below this, there are tabs for "Analysis", "Relationships", "Additional information", "Comments", and "Votes". The "Analysis" tab is selected.
The analysis results are shown in a table with three columns: "Antivirus", "Result", and "Update".

Antivirus	Result	Update
AVG	Generic18.BSSH	20140309
Agnitum	Trojan.Orsam/Giccol39E1oM	20140310
AntiVir	SPR/PWDump.B	20140311
Anti-VL	Trojan(PSWTool-not-a-virus)/Win32.PWDump	20140311
Avas!	Win32.PUP-gen [PUP]	20140311
Baidu-International	HackTool.Win32.PWDump.Ag	20140311
CAT-QuickHeal	HackTool.PWDump (Not a Virus)	20140311
CMC	PSWTool.Win32.PWDump!Q	20140307
ClamAV	Trojan.Pwdump	20140310
Commtouch	W32/Trojan.VJIT-0945	20140311



Anubis: Analyzing Unknown Binaries

<http://anubis.iseclab.org>



Avast! Online Scanner

<http://91.213.143.22>



Malware Protection Center

<https://www.microsoft.com>



ThreatExpert

<http://www.threatexpert.com>



Dr. Web Online Scanners

<http://vms.drweb.com>



Metascan Online

<http://www.metascan-online.com>



Bitdefender QuickScan

<http://quickscan.bitdefender.com>



UploadMalware.com

<http://www.uploadmalware.com>



Online Virus Scanner

<http://www.fortiguard.com>



ThreatAnalyzer

<http://www.threattracksecurity.com>



Ransom Cryptolocker is a ransom-ware that on execution **locks the user's system** thereby leaving the system in an unusable state



It also **encrypts the list of file types** present in the user system



The compromised user has to **pay the attacker** with ransom to unlock the system and to get the files decrypted

Infection and Propagation Vectors



The malware is being propagated via **malicious links in spam e-mails** which leads to pages exploiting common system vulnerabilities



These **exploit pages** will drop Ransom Cryptolocker and other malicious executable files on the affected machine

Characteristics and Symptoms

The contents of the original files are encrypted using **AES Algorithm** with a randomly generated key



Once the system is infected, the malware binary first tries to connect to a hard coded **command and control server** with IP address **184.164.136.134**

01

If this attempt fails, it **generates a domain name** using random domain name algorithm and appends it with domain names such as .org, .net, .co.uk, .info, .com, .biz, and .ru



Encryption Technique

The malware uses an AES algorithm to encrypt the files. The malware first generates a **256 bit AES key** and this will be used to encrypt the files



In order to be able to decrypt the files, the **malware author** needs to know that key



To avoid transmitting the key in clear text, the malware will encrypt it using an **asymmetric key algorithm**, namely the RSA public/private key pair



This encrypted key is then submitted to the **C&C server**





Once the system is compromised, the malware displays the below mentioned **warning** to the user and demand ransom to **decrypt the files**



It maintains the list of files which was encrypted by this malware under the following registry entry

🔑 `HKEY_CURRENT_USER\Software\CryptoLocker\Files`



On execution, this malware binary copies itself to `%AppData%` location and deletes itself using a batch file

🔑 `%AppData%\{2E376276-3A5A-0712-2BE2-FBF2CFF7ECD5}.exe`



Your important files encryption produced on this computer: photos, videos, documents, etc.
If you see this text, but do not see the "CryptoLocker" window, then your antivirus deleted "CryptoLocker" from computer.

If you need your files, you have to recover "CryptoLocker" from the antivirus quarantine, or find a copy of "CryptoLocker" in the Internet and start it again.

You can download "CryptoLocker" from the link given below:

Approximate destruction time of your private key:
3/3/2013 8:15 AM

If the time is finished you are unable to recover files anymore! Simply remove this wallpaper from your desktop.





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Penetration Testing



Scan for suspicious **OPEN PORTS**



Scan for suspicious **RUNNING PROCESSES**



Scan for suspicious **REGISTRY ENTRIES**



Scan for suspicious **DEVICE DRIVERS**
installed on the computer



Scan for suspicious **WINDOWS SERVICES**



Scan for suspicious **STARTUP PROGRAMS**



Scan for suspicious **FILES and FOLDERS**



Scan for suspicious **NETWORK ACTIVITIES**



Scan for suspicious modification to
OPERATING SYSTEM FILES

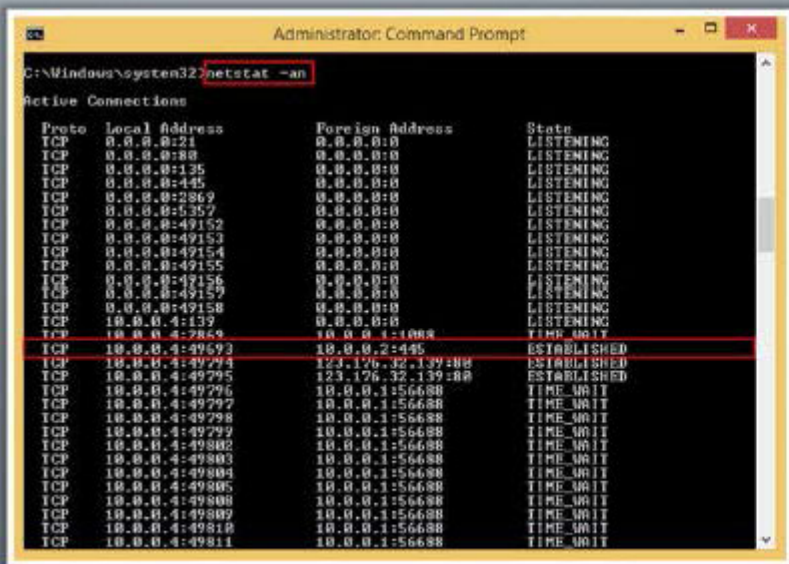


Run Trojan **SCANNER** to detect Trojans



Trojans open **unused ports** in victim machine to connect back to Trojan handlers

Look for the **connection established** to unknown or suspicious IP addresses



```
Administrator: Command Prompt
C:\Windows\system32>netstat -an

Active Connections
Proto Local Address           Foreign Address         State
TCP 0.0.0.0:21               0.0.0.0:0               LISTENING
TCP 0.0.0.0:80               0.0.0.0:0               LISTENING
TCP 0.0.0.0:135              0.0.0.0:0               LISTENING
TCP 0.0.0.0:445              0.0.0.0:0               LISTENING
TCP 0.0.0.0:2869            0.0.0.0:0               LISTENING
TCP 0.0.0.0:5352            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49152           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49153           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49154           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49155           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49156           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49159           0.0.0.0:0               LISTENING
TCP 0.0.0.0:49158           0.0.0.0:0               LISTENING
TCP 10.0.0.4:139            0.0.0.0:0               LISTENING
TCP 10.0.0.4:2869           0.0.0.0:0               LISTENING
TCP 10.0.0.4:2869           10.0.0.1:1000           TIME_WAIT
TCP 10.0.0.4:49593          10.0.0.2:445            ESTABLISHED
TCP 10.0.0.4:49794          129.126.32.139:80       ESTABLISHED
TCP 10.0.0.4:49795          129.126.32.139:80       ESTABLISHED
TCP 10.0.0.4:49796          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49797          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49798          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49799          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49800          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49801          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49802          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49803          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49804          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49805          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49806          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49807          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49808          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49809          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49810          10.0.0.1:5668           TIME_WAIT
TCP 10.0.0.4:49811          10.0.0.1:5668           TIME_WAIT
```

Type **netstat -an**
in command prompt



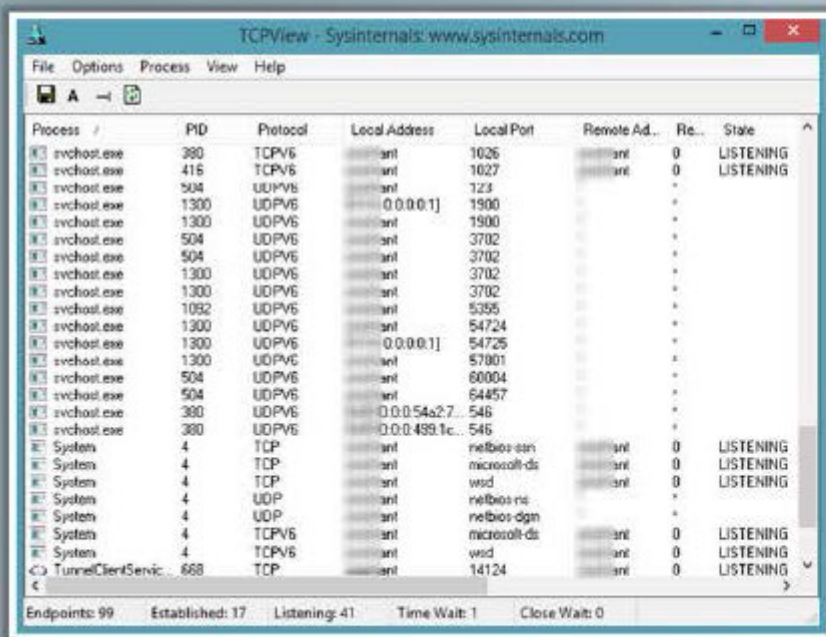
System Administrator

TCPView

TCPView show detailed listings of all **TCP** and **UDP endpoints** on your system, including the local and remote addresses and state of **TCP connections**

CurrPorts

CurrPorts is **network monitoring** software that displays the list of all currently opened **TCP/IP** and **UDP** ports on your local computer

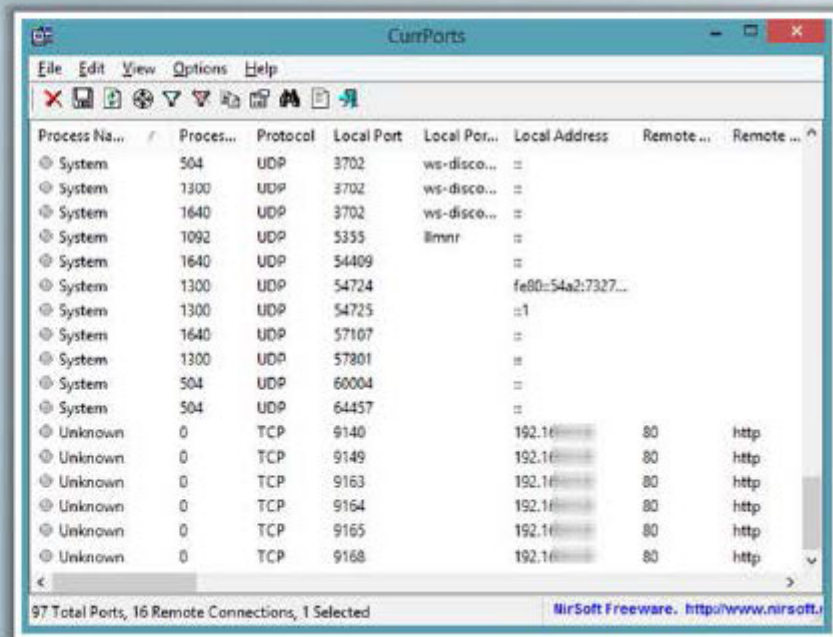


TCPView - Sysinternals: www.sysinternals.com

Process	PID	Protocol	Local Address	Local Port	Remote Ad...	Re...	State
svchost.exe	380	TCPV6	ant	1026	ant	0	LISTENING
svchost.exe	416	TCPV6	ant	1027	ant	0	LISTENING
svchost.exe	504	UDPV6	ant	123		*	
svchost.exe	1300	UDPV6	0.0.0.0::1	1900		*	
svchost.exe	1300	UDPV6	ant	1900		*	
svchost.exe	504	UDPV6	ant	3702		*	
svchost.exe	504	UDPV6	ant	3702		*	
svchost.exe	1300	UDPV6	ant	3702		*	
svchost.exe	1300	UDPV6	ant	3702		*	
svchost.exe	1092	UDPV6	ant	5355		*	
svchost.exe	1300	UDPV6	ant	54724		*	
svchost.exe	1300	UDPV6	0.0.0.0::1	54725		*	
svchost.exe	1300	UDPV6	ant	57801		*	
svchost.exe	504	UDPV6	ant	60004		*	
svchost.exe	504	UDPV6	ant	64457		*	
svchost.exe	380	UDPV6	0.0.0.54a27...	546		*	
svchost.exe	380	UDPV6	0.0.0.489.1c...	546		*	
System	4	TCP	ant	netbios-ssn	ant	0	LISTENING
System	4	TCP	ant	microsoft-ds	ant	0	LISTENING
System	4	TCP	ant	wsd	ant	0	LISTENING
System	4	UDP	ant	netbios-ns		*	
System	4	UDP	ant	netbios-dgm		*	
System	4	TCPV6	ant	microsoft-ds	ant	0	LISTENING
System	4	TCPV6	ant	wsd	ant	0	LISTENING
System	4	TCP	ant	14124	ant	0	LISTENING

Endpoints: 99 Established: 17 Listening: 41 Time Wait: 1 Close Wait: 0

<http://technet.microsoft.com>



CurrPorts

Process Na...	Proces...	Protocol	Local Port	Local Port...	Local Address	Remote ...	Remote ...
System	504	UDP	3702	ws-disco...	::		
System	1300	UDP	3702	ws-disco...	::		
System	1640	UDP	3702	ws-disco...	::		
System	1092	UDP	5355	limnr	::		
System	1640	UDP	54409		::		
System	1300	UDP	54724		fe80::54a2:7327...		
System	1300	UDP	54725		::1		
System	1640	UDP	57107		::		
System	1300	UDP	57801		::		
System	504	UDP	60004		::		
System	504	UDP	64457		::		
Unknown	0	TCP	9140		192.168.1.100	80	http
Unknown	0	TCP	9149		192.168.1.100	80	http
Unknown	0	TCP	9163		192.168.1.100	80	http
Unknown	0	TCP	9164		192.168.1.100	80	http
Unknown	0	TCP	9165		192.168.1.100	80	http
Unknown	0	TCP	9168		192.168.1.100	80	http

97 Total Ports, 16 Remote Connections, 1 Selected

MirSoft Freeware. <http://www.nirsoft.net>

<http://www.nirsoft.net>

01

Trojans camouflage themselves as **genuine Windows services** or hide their processes to avoid detection

Some Trojans use PEs (**Portable Executable**) to inject into various processes (such as explorer.exe or web browsers)

02

03

Processes are visible but looks like a legitimate processes and also helps **bypass desktop firewalls**

Trojans can also use **rootkit** methods to hide their processes

04

05

Use **process monitoring** tools to detect hidden Trojans and backdoors

Process Monitor

Process Monitor is a monitoring tool for Windows that **shows file system, registry, and process/thread activity**



Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time	Process Name	PID	Operation	Path	Result	Detail
10:01...	Explorer EXE	1728	QueryStandard...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 1.0...
10:01...	Explorer EXE	1728	CreateFileMap...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	SyncType: SyncTy...
10:01...	Explorer EXE	1728	QueryStandard...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 1.0...
10:01...	Explorer EXE	1728	QueryStandard...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 1.0...
10:01...	Explorer EXE	1728	CloseFile	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 1.0...
10:01...	Explorer EXE	1728	QueryStandard...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 8.1...
10:01...	Explorer EXE	1728	QueryStandard...	C:\Users\admin\AppData\Local\Micros...	SUCCESS	AllocationSize: 1.0...
10:01...	Explorer EXE	1728	CreateFile	C:\Users\admin\Desktop	SUCCESS	Desired Access: R...
10:01...	Explorer EXE	1728	QueryMiniMet...	C:\Users\admin\Desktop	INVALID PA...	
10:01...	Explorer EXE	1728	QueryDirectory	C:\Users\admin\Desktop	SUCCESS	0: 1: 2: deskto...
10:01...	Explorer EXE	1728	CreateFile	C:\Users\Public\Desktop	SUCCESS	Desired Access: R...
10:01...	Explorer EXE	1728	QueryRemotePr...	C:\Users\Public\Desktop	INVALID PA...	
10:01...	Explorer EXE	1728	QueryDirectory	C:\Users\Public\Desktop	SUCCESS	
10:01...	Explorer EXE	1728	RegQueryKey	HKLM	SUCCESS	
10:01...	Explorer EXE	1728	RegOpenKey	HKLM\Software\Microsoft\Windows\IC...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
10:01...	Explorer EXE	1728	RegEnumKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	

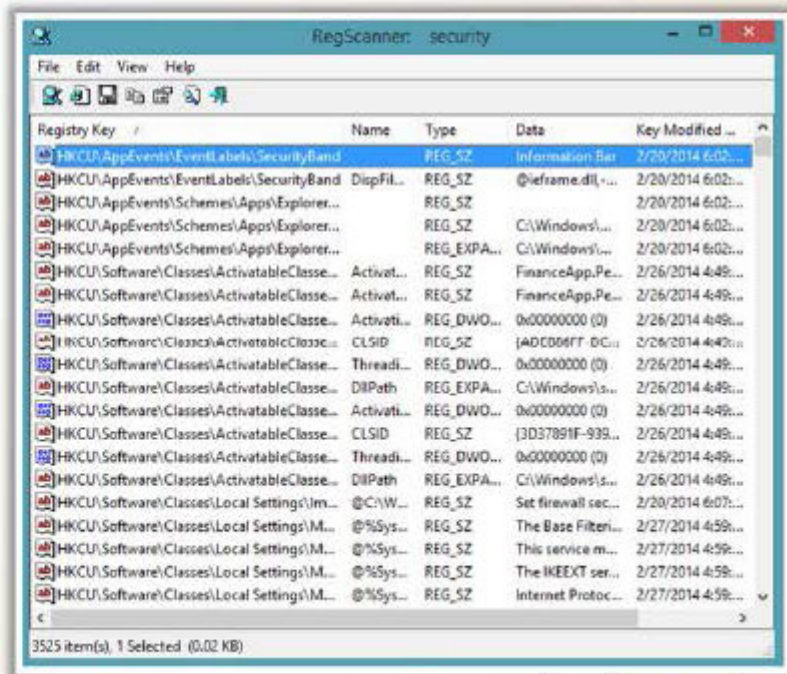
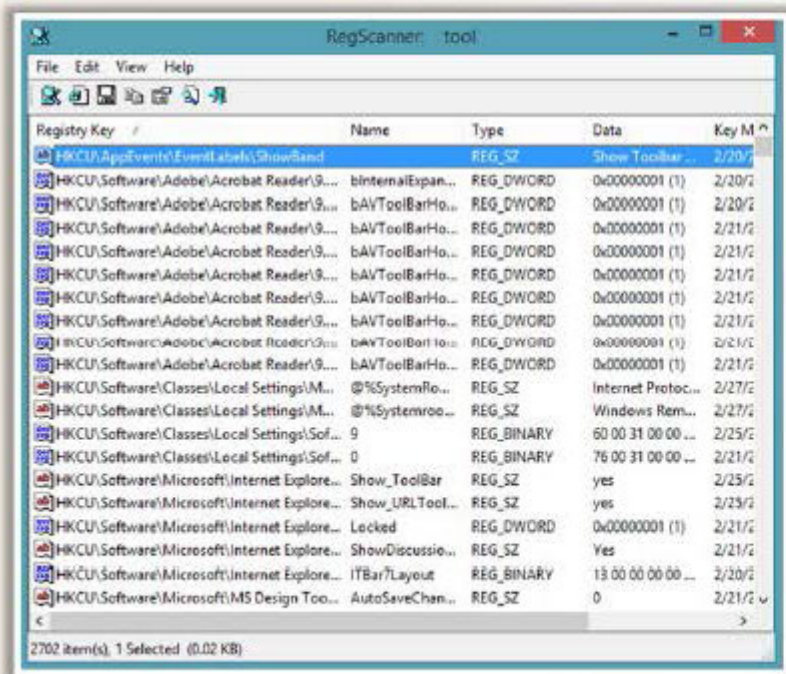
Showing 27,395 of 116,315 events (23%) Backed by virtual memory

Index: 4. Name: 12...
Index: 5. Name: 14...
Index: 6. Name: 14...

0: ...
1: ...
2: desktop.ini
3: New Text Document.txt
4: ProcessMonitor
5: stego.png
6: Thumbs.db
7: Trojan.jar



RegScanner allows you to scan the Registry, **find the desired Registry values** that match to the specified search criteria, and display them in one list



Check startup
program entries
in the registry

Details are covered in next slide



Check device
drivers automati-
cally loaded

`C:\Windows\System32\drivers`



Check
`boot.ini`

Check `boot.ini` or `bcd` (bootmgr) entries



Check Windows
services
automatic started

Go to **Run** → Type `services.msc` → Sort by
Startup Type



Check startup
folder

`C:\ProgramData\Microsoft\Windows\Start
Menu\Programs\Startup`
`C:\Users\ (User-Name) \AppData\Roaming\
Microsoft\Windows\Start Menu\Programs\Startup`

Security AutoRun displays the list of all applications that are loaded automatically when Windows starts up

The screenshot shows the Security AutoRun application window. On the left is a tree view with categories like 'Local Machine', 'WinLogon', 'Startup Folder', 'Startup Services', 'Driver List', 'Know DLLs', 'AppInit DLLs', 'Scheduled Tasks', and 'Ini Startup'. The 'Local Machine' category is expanded, showing sub-items like 'Run (2)', 'Run Once (0)', 'Run OnceEx (0)', 'RunServicesOnce (0)', 'Policies (0)', 'ShellObject (1)', and 'Installed Components (5)'. The 'Run' sub-item is selected, displaying a list of services in the main pane. The list has columns for 'Service Name', 'Description', 'Status', and 'Path'. The status of services varies, with some being 'Stopped', 'Running', or 'Not Installed'. The path column shows the file location for each service, mostly within the Windows system directory. A blue ribbon graphic is overlaid on the top of the window, pointing towards the 'Run' sub-item in the tree view.

Service Name	Description	Status	Path
COMSysApp	COM+ System Application	Stopped	C:\Windows\system32\lsass.exe /ProcessId:0x24B3F...
defragsvc	Optimize drives	Stopped	C:\Windows\system32\svchost.exe -k defragsvc
Fax	Fax	Stopped	C:\Windows\system32\svchost.exe -k defragsvc
gupdate	Google Update Service (gupdate)	Stopped	"C:\Program Files (x86)\Google\Update\GoogleUpdate.e...
gupdate	Google Update Service (gupdate)	Stopped	"C:\Program Files (x86)\Google\Update\GoogleUpdate.e...
IEEtrCollectorSer...	Internet Explorer ETW Collector Se...	Stopped	C:\Windows\system32\IEEtrCollector.exe /f
MSDTC	Distributed Transaction Coordinator	Stopped	C:\Windows\system32\lsass.exe /f
msiserver	Windows Installer	Stopped	C:\Windows\system32\msiexec.exe /f
nvsvc	NVIDIA Display Driver Service	Running	"C:\Windows\system32\nvsvc.exe"
nvupdateService	NVIDIA Update Service Daemon	Running	C:\Windows\system32\NVIDIA Corporation\NVIDIA Upd...
ose	Office Source Engine	Stopped	"C:\Program Files (x86)\Common Files\Microsoft Shared\...
osppsvc	Office Software Protection Platform	Running	"C:\Program Files\Common Files\Microsoft Shared\Office...
Perfhost	Performance Counter DLL Host	Stopped	C:\Windows\System64\perfhost.exe
rpsvc	Remote Packet Capture Protocol v....	Stopped	"C:\Program Files (x86)\WinPcap\rpsvc.exe" -d -f "C:\P...
RpcLocator	Remote Procedure Call (RPC) Locator	Stopped	C:\Windows\system32\locator.exe
smphost	Microsoft Storage Spaces SPM	Stopped	C:\Windows\System32\svchost.exe -k smphost
SNMPTRAP	SNMP Trap	Stopped	C:\Windows\System32\snmptrap.exe
Spooler	Print Spooler	Running	C:\Windows\System32\spooler.exe
svchost	Software Protection	Stopped	-
Stereo Service	NVIDIA Stereoscopic 3D Driver Ser...	Running	"C:\Program Files (x86)\NVIDIA Corporation\3D Vision\rv...
tschost	Windows Image Acquisition (WIA)	Stopped	C:\Windows\system32\svchost.exe -k imgsvc
svcpv	Microsoft Software Shadow Copy P...	Stopped	C:\Windows\System32\svchost.exe -k svcpv
Windows Modules Installer	Windows Modules Installer	Stopped	-
Wscntfy	Interactive Services Detection	Stopped	C:\Windows\system32\Wscntfy.exe



Trojans connect **back to handlers** and send confidential information to attackers

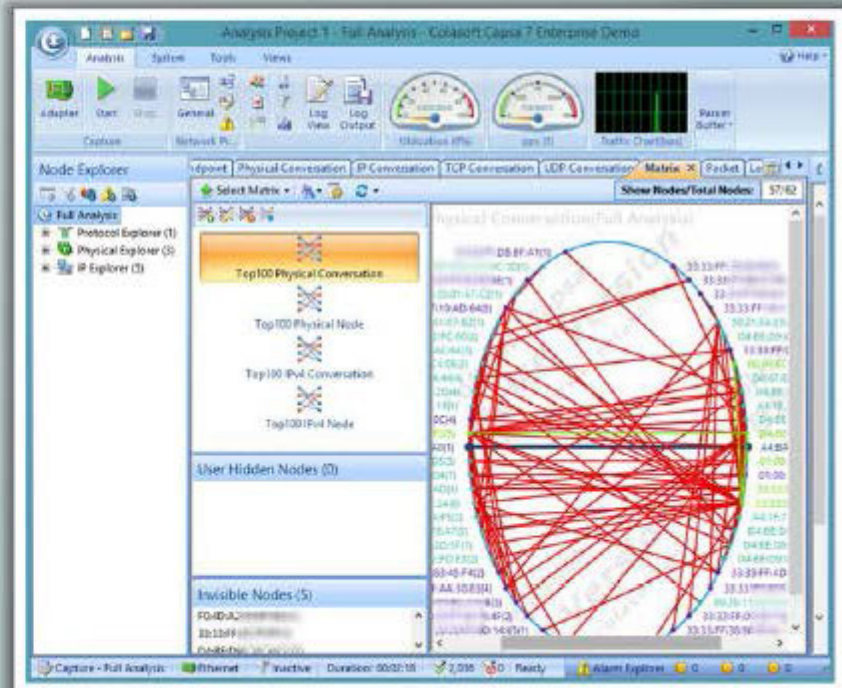
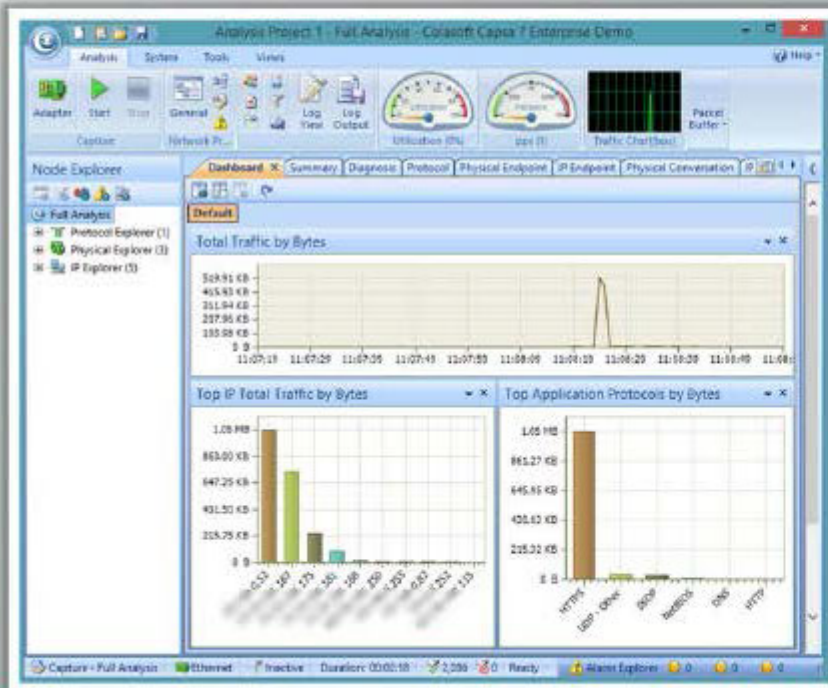


Use network scanners and packet sniffers to monitor **network traffic** going to malicious remote addresses



Run tools such as **Capsa** to monitor network traffic and look for suspicious activities sent over the web

Capsa is an intuitive network analyzer, which provides detailed information to help check if there are any **Trojan activities on a network**



Scanning

Once a virus has been detected, it is possible to write scanning programs that look for signature string characteristics of the virus



Integrity Checking

Integrity checking products work by reading the entire disk and recording integrity data that acts as a signature for the files and system sectors



Interception

The interceptor monitors the operating system requests that are written to the disk



Code Emulation



- In code emulation techniques, the **anti-virus executes the malicious code** inside a virtual machine to simulate CPU and memory activities
- This technique is considered very effective in dealing with **encrypted** and **polymorphic viruses** if the virtual machine mimics the real machine

Heuristic Analysis



- Heuristic analysis can be **static** or **dynamic**
- In static analysis the **anti-virus analyses the file format** and code structure to determine if the code is viral
- In dynamic analysis the **anti-virus performs a code emulation** of the suspicious code to determine if the code is viral



Introduction to Malware



Trojan Concepts



Virus and Worm Concepts



Malware Reverse Engineering



Malware Detection



Counter- measures



Anti-Malware Software



Penetration Testing



Avoid opening **email attachments** received from unknown senders



Block all **unnecessary ports** at the host and firewall



Avoid accepting the programs transferred by **instant messaging**



Harden weak, **default configuration** settings and disable **unused functionality** including protocols and services



Monitor the **internal network traffic** for odd ports or encrypted traffic



Avoid downloading and executing applications from **untrusted sources**



Install patches and **security updates** for the operating systems and applications



Scan CDs and DVDs with **antivirus software** before using



Restrict permissions within the **desktop environment** to prevent malicious applications installation



Avoid typing the commands blindly and implementing **pre-fabricated programs or scripts**



Manage local workstation **file integrity** through checksums, auditing, and port scanning



Run **host-based antivirus**, firewall, and intrusion detection software



Most commercial **anti-virus products** can automatically scan and detect **backdoor programs** before they can cause damage



Educate users not to install applications downloaded from **untrusted Internet sites** and **email attachments**



Use **anti-virus tools** such as McAfee, Norton, etc. to detect and eliminate backdoors

Install **anti-virus** software that detects and removes infections as they appear

01



Pay attention to the **instructions** while downloading files or any programs from the Internet

03

02

Generate an **anti-virus policy** for safe computing and distribute it to the staff

Avoid opening the attachments received from an **unknown sender** as viruses spread via e-mail attachments

05

04

Update the anti-virus software regularly

Schedule **regular scans** for all drives after the installation of anti-virus software

07

06

Possibility of virus infection may corrupt data, thus regularly maintain **data back up**

08

Do not accept disks or programs without checking them first using a **current version** of an anti-virus program



Ensure the **executable code** sent to the organization is approved

1

Do not boot the machine with **infected** bootable system disk

2

Know about the **latest virus** threats

3

Check the **DVDs** and **CDs** for virus infection

4

Ensure the **pop-up blocker** is turned on and use an Internet firewall

5

6

Run disk clean up, registry scanner and **defragmentation** once a week

7

Turn on the **firewall** if the OS used is Windows XP

8

Run **anti-spyware** or **adware** once in a week

9

Do not open the files with more than one **file type extension**

10

Be cautious with the files being sent through the **instant messenger**



Introduction to Malware



Trojan Concepts



Virus and Worm Concepts



Malware Reverse Engineering



Malware Detection



Counter- measures



Anti-Malware Software



Penetration Testing



Anti Malware BOClean

<http://www.comodo.com>



Anti Hacker

<http://www.hide-my-ip.com>



XoftSpySE

<http://www.paretologic.com>



SPYWAREfighter

<http://www.spamfighter.com>



Malwarebytes Anti-Malware Premium

<http://www.malwarebytes.org>



SUPERAntiSpyware

<http://www.superantispyware.com>



Trojan Remover

<http://www.simplysup.com>



Twister Antivirus

<http://www.filseclab.com>



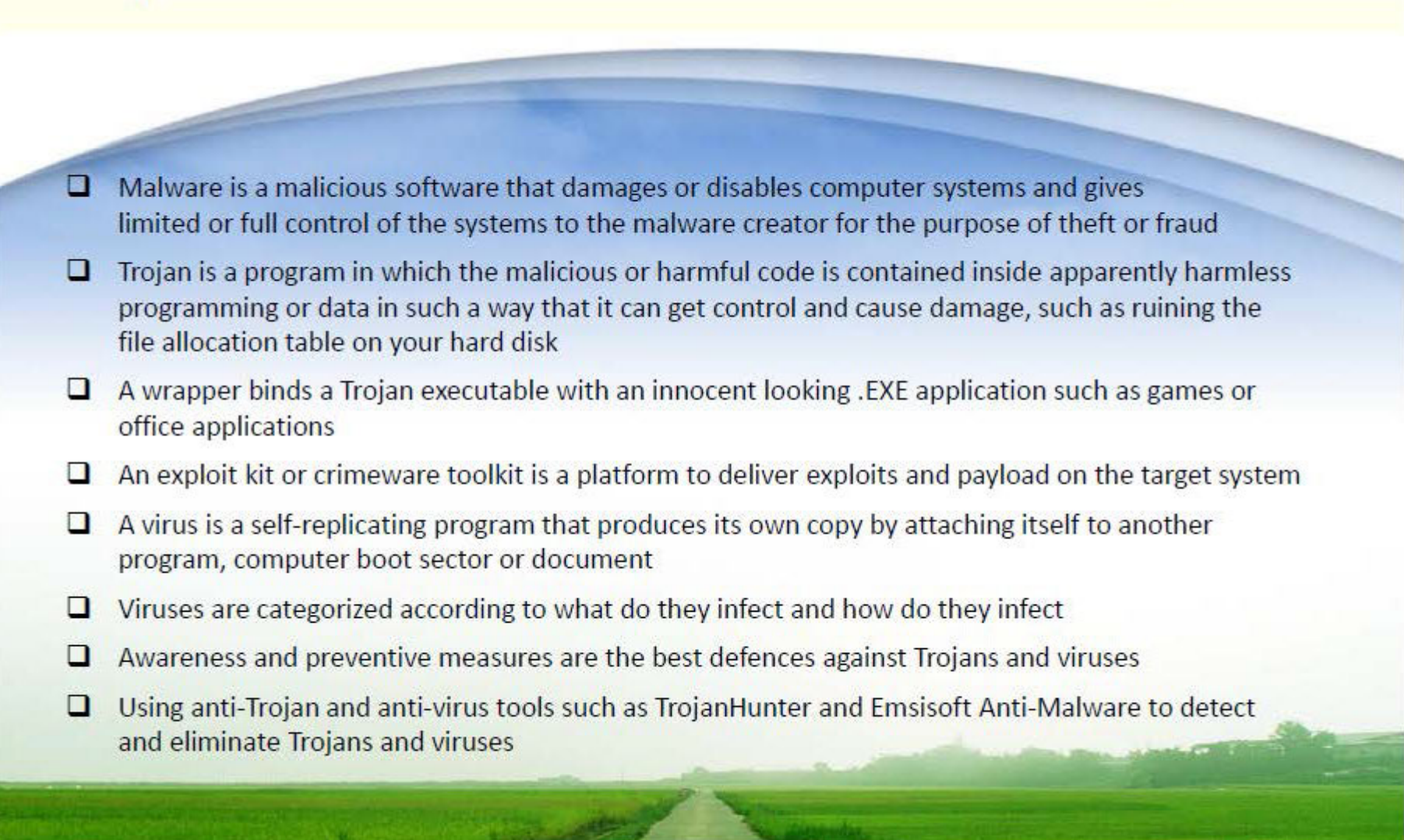
STOPzilla AntiMalware

<http://www.stopzilla.com>



ZeroSpyware

<http://www.fbmssoftware.com>

- 
- ☐ Malware is a malicious software that damages or disables computer systems and gives limited or full control of the systems to the malware creator for the purpose of theft or fraud
 - ☐ Trojan is a program in which the malicious or harmful code is contained inside apparently harmless programming or data in such a way that it can get control and cause damage, such as ruining the file allocation table on your hard disk
 - ☐ A wrapper binds a Trojan executable with an innocent looking .EXE application such as games or office applications
 - ☐ An exploit kit or crimeware toolkit is a platform to deliver exploits and payload on the target system
 - ☐ A virus is a self-replicating program that produces its own copy by attaching itself to another program, computer boot sector or document
 - ☐ Viruses are categorized according to what do they infect and how do they infect
 - ☐ Awareness and preventive measures are the best defences against Trojans and viruses
 - ☐ Using anti-Trojan and anti-virus tools such as TrojanHunter and Emsisoft Anti-Malware to detect and eliminate Trojans and viruses